

Student ID Management System

Project Name: Student ID Management System

Student Name: Aryan Gupta

Registration Number: 26MIB10002

Language Used: Python

Project Type: Basic Python Project

About the Project

The Student ID Management System is a simple Python-based project designed to manage basic student information in one place. It allows the user to add student details, search for a student, update information, delete records, and display a student ID card. It also includes attendance, marks, grades, and fee status. The project uses basic Python concepts and is suitable for beginners.

Features

1. **Add Student:** Add a student's ID, name, age, gender, course, branch, year, phone number, and email.
2. **Show All Students:** Display the basic information of all students stored in the system.
3. **Search Student:** Find a student using their Student ID.
4. **Update Student:** Update an existing student's details.
5. **Delete Student:** Remove a student's record.
6. **Student ID Card:** Display a simple digital student ID card.
7. **Attendance:** Add or update attendance percentage.
8. **Marks:** Enter the student's marks percentage.
9. **Grade:** Calculate a grade based on marks.
10. **Fee Status:** Set fees as Paid or Not Paid.
11. **Student Report:** Display the student's basic and academic information.

Python Concepts Used

Variables, lists, dictionaries, functions, if-elif-else statements, for loops, while loops, user input, searching, updating and deleting list data, basic calculations, and string formatting. No external Python libraries are required.

How the Project Works

When the program starts, a menu is displayed. The user enters the number of the required option. The program then performs the selected operation using the corresponding function.

```
=====
      STUDENT ID MANAGEMENT
=====
1. Add Student
2. Show All Students
3. Search Student
4. Update Student
5. Delete Student
6. Show Student ID Card
```

7. Update Attendance
8. Enter Marks
9. Show Grade
10. Fee Status
11. Student Report
12. Exit

How to Run the Project

1. Install Python on your computer.
2. Open the project in VS Code or another Python-supported editor.
3. Save the program as **main.py**.
4. Run the program using the editor's Run Python File option or the terminal:

```
python main.py
```

Project Structure

```
Student_ID_Management_System/  
■■■ main.py  
■■■ README.md
```

Example

A student can be added by entering details such as Student ID, name, age, gender, course, branch, year, phone number, and email. The record can then be searched using the Student ID.

Limitations

Student data is stored temporarily in the program. Records will be lost when the program is closed because the project does not use a database or file storage system.

Future Improvements

Possible improvements include database connectivity, a login system, student photos on ID cards, file-based data storage, a better ID card design, password protection, exporting reports, and a graphical user interface (GUI).

Conclusion

The Student ID Management System demonstrates how basic Python programming concepts can be used to create a useful management application. It helps in understanding functions, lists, dictionaries, loops, conditions, and user input while providing simple student record management features.

Student Details

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